HashMol3D: A Canonical Identifier for Quantized Molecular Geometries

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Abstract

Computational chemistry datasets require geometry identifiers for indexing, exact deduplication, and cache lookup, whereas canonical SMILES and InChI primarily encode connectivity. We present HashMol3D, which maps an element-labeled three-dimensional geometry to a short identifier. The default method (tag `F`) constructs a deterministic frame from an atomic-number-weighted gyration tensor. Separated principal moments use their eigenvectors; degenerate moments use intrinsic point or line coordinates, or canonical atom anchors selected from invariant integer keys. Tied anchors are all evaluated, so no random displacement is introduced. The resulting sorted, quantized coordinate rows require $O(N)$ memory. A complete canonicalized distance-matrix method (tag `C`) remains available explicitly and is also the deterministic fallback for ill-conditioned frames; its fixed-budget fallback is tagged `W` and is invariant but incomplete. Tests verify the path-specific claims and confirm that homometric counterexamples are separated. Across 1,528,722 QM9 and MD17 geometries, 1,528,720 used `F`, two used the complete `C` fallback, none used `W`, and all 128-bit digests were distinct at the default 10^{-4} Å grid. A six-grid corpus sweep found no repeated full digest at 10^{-4} or 10^{-5} Å, 25 excess repetitions at 10^{-3} Å, and rapid coalescence of dense trajectories on coarser grids. The 459,630 anchored cases include 450,622 benzene frames. Short-prefix collision counts follow the exact occupancy model. Coordinate-noise tests confirm that both exact representations are discontinuous at their quantization boundaries. HashMol3D is therefore intended for exact matching on a recorded method and grid, not tolerance-based structural comparison.

1 Introduction

Computational chemistry produces large collections of three-dimensional (3D) structures: conformer ensembles, molecular-dynamics trajectories, and quantum-chemistry (QC) or machine-learning (ML) datasets. These collections often need exact geometry keys for indexing, provenance, and avoiding repeated calculations.

Canonical SMILES [1, 2] and the IUPAC InChI [3] canonicalize a molecular *graph*: an atom-and-bond connectivity table, optionally augmented with stereochemical layers. Because these identifiers omit conformer coordinates, distinct 3D geometries can have the same string. Raw coordinate files have the opposite problem: translation, rotation, atom reordering, or floating-point noise changes their byte representation. Pairwise root-mean-square deviation (RMSD) after optimal alignment [4] requires $O(M^2)$ comparisons for M structures, with an alignment over N atoms for each comparison, and does not provide a portable key.

HashMol3D assigns a short key to an element-labeled geometry. Its equivalence target follows symmetries of the clamped-nuclei electronic Hamiltonian: rigid motions, spatial inversion, and relabeling of identical nuclei. Because the input contains atomic numbers but not nuclear masses, the identifier also merges isotopologues. These choices are explicit properties of the identifier; they are not a claim that the encoded operations exhaust all spectrum-preserving transformations.

This paper defines the three possible descriptor paths—`F`, `C`, and `W`—and their distinct guarantees (Section 3). We prove the frame- and canonical-path properties (Section 4) and test the implementation with adversarial counterexamples, public datasets, and coordinate perturbations (Section 5). Detailed parameter studies are reported in the supplementary material and are not used to claim a universal optimal precision.

2 Background and Related Work

Canonical line notations. SMILES provides a compact notation for molecular structure [1]; canonical SMILES imposes an atom order derived from a Morgan-style invariant [2, 5]. InChI defines a layered standardized identifier and the fixed-length InChIKey, a truncated and base-encoded SHA-256 hash of its layers, for database and web search [3]. HashMol3D likewise hashes a canonical description, but describes 3D geometry rather than 2D connectivity.

Canonicalization and graph isomorphism. Producing a canonical atom order is related to graph canonical labeling [6]. Greedy refinement schemes such as Morgan's algorithm [5] leave ties in symmetric structures. HashMol3D resolves these ties with Weisfeiler–Leman color refinement [7] followed by an individualization-refinement search with a lexicographically minimal target, the paradigm of nauty [6], but applies it to the precision-rounded *distance matrix* rather than to an adjacency matrix. For generic geometries, distances usually distinguish all atoms in one refinement round. Symmetric structures require branching over the remaining orbits. Earlier versions instead hashed sorted distance multisets, which are invariant but not complete (Section 4).

Invariant 3D representations in machine learning. Many 3D ML descriptors are invariant under rotation, translation, and atom permutation. Coulomb-matrix variants encode nuclear charges and inverse distances [8]; Bag of Bonds groups terms by element pair [9]; SOAP compares local atomic densities [10]; symmetry functions encode radial and angular environments [11, 12]; and FCHL and MBTR use distributions of distances and angles [13, 14]. These continuous representations target regression or similarity. HashMol3D instead quantizes and hashes a geometry for exact matching. Distance multisets and invariant atomic environments can be nonunique [15, 16]; the canonical matrix removes this ambiguity at the working precision. A canonical frame retains the full element-labeled coordinate rows and likewise avoids the multiset reduction. We compare these constructions with Coulomb-matrix eigenvalues and a naive principal-axis descriptor in Section 5.1. The frame method of Section 3.4 resolves degenerate axes deterministically and falls back to the canonical matrix only when no well-conditioned anchored frame is available.

Fingerprints and hashing. Extended-connectivity fingerprints hash local 2D substructures into fixed-size bit vectors for similarity search [17]. USR and USRCAT use distance distributions to form rotation- and translation-invariant 3D shape vectors [18, 19]. Open-source toolkits such as jCompoundMapper implement a range of 2D and 3D fingerprint encodings [20], and PubChem3D provides precomputed conformer models with shape- and feature-based 3D similarity across a large public database [21]. These methods target similarity rather than exact geometric identity.

Distance geometry and the homometric problem. The full interatomic distance matrix determines a point configuration up to isometry and reflection [22–24]. This result underlies the completeness claim and explains why a distance-based identifier cannot distinguish enantiomers. Uniqueness is lost when the matrix is reduced to an unordered multiset. Reconstruction from such data is the turnpike or beltway problem [25, 26]; the related unassigned distance-geometry problem is studied in Ref. 27. Although generic configurations are determined by their distance distributions, exceptions exist [28]. Configurations sharing a distance multiset are called *homometric*, a phenomenon first identified in X-ray crystallography as an ambiguity of the Patterson function [29–31]; the classical minimal example is Bloom's pair of six-mark rulers [32]. Canonicalizing the labeled matrix retains the assignment of distances to atom pairs and therefore separates noncongruent homometric configurations (Section 4). Recent work in computational geometry constructs complete, continuous isometry invariants for unlabeled point clouds [33, 34]. Related invariants and metrics have been applied to near-duplicate detection in molecular and materials databases [35]. This literature addresses the harder geometric problem of measuring similarity continuously and, in fixed dimension, provides polynomial-time constructions. HashMol3D does not introduce a new continuous isometry invariant. Its contribution is a software and serialization standard for an element-labeled, quantized geometry: it specifies the grid, canonical byte representation, versioning, fallback behavior, and digest namespace needed for exact database keys. Continuous metrics and discrete identifiers therefore serve complementary purposes. The former compare noisy structures; the latter support exact lookup after an application has selected a quantization policy.

Cryptographic hash. HashMol3D uses SHA-256 [36]. The user may select the digest length; the default is 32 hexadecimal characters (128 bits).

3 Method

3.1 Input and definition of equivalence

The input comprises atomic numbers $Z = (Z_1, \dots, Z_N)$, $Z_i \in \{1, \dots, 118\}$, and Cartesian coordinates $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbb{R}^{N\times 3}$ in Ångström. The target equivalence allows a joint permutation of atoms, translation, and any orthogonal transformation, including reflection. Thus, for some permutation $\pi \in S_N$, matrix $R \in O(3)$, and translation $\mathbf{t}$, equivalent inputs satisfy

$$Z_i = Z'_{\pi(i)}, \qquad \mathbf{x}'_{\pi(i)} = R\mathbf{x}_i + \mathbf{t}.$$

The input has no isotope field, so isotopologues are indistinguishable.

The grid spacing is restricted to $\varepsilon = 10^{-p}$ Å for an integer $p \geq 0$. Other values and grids coarser than 1 Å are rejected. The default frame method quantizes centered coordinates in a deterministic frame; the retained canonical method quantizes pairwise distances. In both cases, the output is an exact key for a discrete representation. It is not an "equal within ε" test on Cartesian coordinates.

3.2 Path overview: what `F`, `C`, and `W` mean

The final field of every descriptor begins with a one-letter path tag. The tag is part of the hashed payload and identifies both the representation and its guarantees. Table 1 is normative.

Figure 1 shows the path decision before giving the algorithms in detail. The corpus, collision, and primary perturbation studies use the default frame method; the canonical option is retained in comparison artifacts and supplementary analyses.

tag	when returned	stored body	meaning of equality
F	the default frame request finds a principal, intrinsic, or well-conditioned atom-anchored frame within 10^4 candidates	sorted element-labeled, quantized canonical-frame coordinate rows	complete certificate for equality of the element-labeled coordinate-row multisets on that frame grid
C	the explicit canonical request, or a frame fallback, finishes its tie-breaking search within 10^4 states	element labels and the upper triangle of the quantized distance matrix in canonical atom order	complete certificate for isomorphism of the element-labeled integer matrix (Z, Q)
W	the canonical search exceeds 10^4 states	sorted multiset of stable per-atom refinement signatures	invariant and deterministic, but not complete; distinct matrices can share a descriptor

Table 1: The three descriptor paths. Completeness is path specific: F certifies its quantized canonical-frame rows, whereas C certifies (Z, Q). A frame request can return C or W only after deterministic canonical fallback.

3.3 The canonical descriptor

The C path represents a geometry by its element-labeled, quantized distance matrix in an order determined by that matrix. The construction has three operations.

1. Quantize the distances. For each unordered atom pair $\{i, j\}$, calculate $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ and express it as an integer number of grid intervals:

$$q_{ij} \;=\; \mathrm{rint}\big(d_{ij} \cdot 10^p\big), \qquad p = -\log_{10} \varepsilon \in \mathbb{Z}_{\geq 0},$$

where `rint` denotes round-half-to-even. The symmetric integer matrix $\mathbf{Q} = (q_{ij})$ has a zero diagonal. It is unchanged by translation, rotation, or reflection, but its row order still follows the input atom order.

2. Refine atom classes. Here a *color* is an integer class label, not a chemical or graphical property. Initially, atoms of the same element have the same color. At iteration t, atom i is assigned the signature

$$s_i^{(t)} = \Big(c_i^{(t)},\ \mathrm{sort}\{(c_j^{(t)}, q_{ij}) : j \neq i\}\Big)\,.$$

The signatures are sorted lexicographically and replaced by consecutive integer colors. This operation is repeated until no class splits further [7]. Because the previous color is the first component of $s_i^{(t)}$, classes can split but cannot merge. A generic geometry often ends with one atom per class. Symmetry can leave ties; for example, ideal benzene retains one carbon class and one hydrogen class. This step is the one-dimensional Weisfeiler–Leman refinement used in the implementation.

3. Resolve every remaining tie. If every stable class contains one atom, sorting by color fixes the atom order. Otherwise, the algorithm selects the smallest tied class. Each member is, in turn, given a temporary unique color, after which Step 2 is repeated. The procedure recurses until every class is a singleton, as in canonical graph labeling [6]. Every leaf gives a candidate atom order. The algorithm retains the order whose row-major upper triangle of $\mathbf{Q}$ is lexicographically smallest. Thus, the result does not depend on which input index happened to identify an atom. A generic

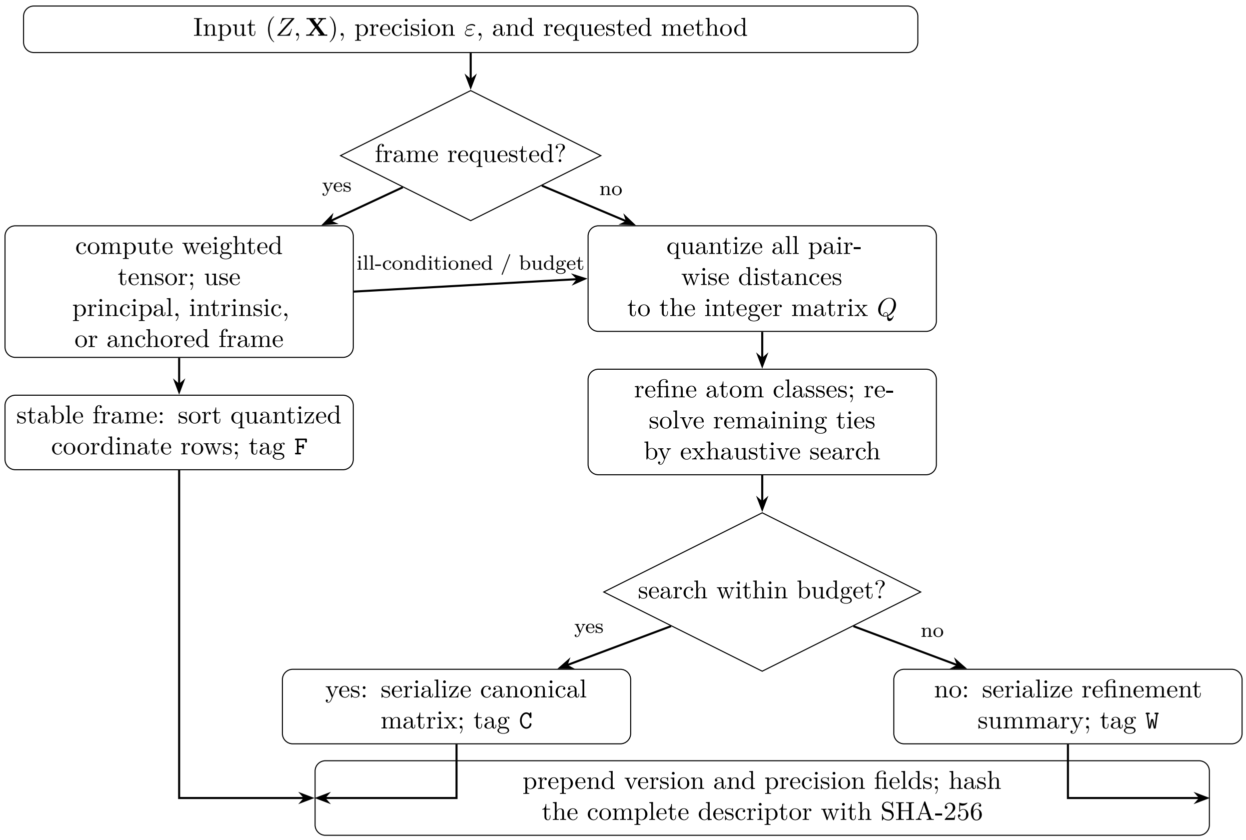

Figure 1: Path selection and serialization. The frame branch is the default; "refine atom classes" means the integer partition-refinement procedure defined in Section 3.3; "exhaustive search" means that all unresolved choices are explored up to the fixed node budget. No acronym in the figure is needed to follow the decision.

geometry requires one leaf; ideal benzene requires 12, a regular tetrahedral cluster 24, and an ideal monoelemental n-membered ring $2n$.

Water illustrates the complete `C` path. At $\varepsilon = 10^{-4}$ Å, the two H–O distances both give 9575 and the H–H distance gives 15144. With initial colors $c_{\mathrm{H}} = 0$ and $c_{\mathrm{O}} = 1$, each hydrogen has signature $[0;\, (0, 15144), (1, 9575)]$, whereas oxygen has $[1;\, (0, 9575), (0, 9575)]$. Refinement therefore separates oxygen but leaves the two equivalent hydrogens tied. Step 3 explores the two hydrogen choices; both give the same upper triangle. The serialized descriptor is

```
V:6-FRAME-SHA256|P:1.0e-04|Z:1,1,8|C:15144,9575,9575
```

It hashes to a suffix beginning `59781410`. The `V` field fixes the descriptor version, `P` records the grid, `Z` gives the atomic numbers in canonical order, and `C` contains the corresponding upper triangle.

The `W` fallback. The exact tie search is limited to 10^4 visited partition states. Exceeding this budget is itself permutation invariant because the search tree is defined only by $(Z, \mathbf{Q})$. The implementation then serializes the sorted multiset of stable per-atom signatures and marks the field `W`. For example, 12 carbon atoms placed within a box much smaller than a 1 Å grid all have $q_{ij} = 0$. Their tie search exceeds the budget and returns a `W` descriptor. This fallback preserves the invariances in Proposition 1, but it is not a complete invariant: distinct quantized distance matrices can, in principle, share the same stable-signature multiset. The separate `C` and `W` tags prevent cross-path collisions and make this loss of completeness visible to the caller.

3.4 Default deterministic frame descriptor

The default `method="frame"` path avoids the $O(N^2)$ distance matrix. It uses a different quantization from the retained canonical method and therefore occupies the separate `F` namespace. The construction begins with the atomic-number-weighted centroid and gyration tensor,

$$\mathbf{c} = \frac{\sum_i Z_i \mathbf{x}_i}{\sum_i Z_i}, \qquad T_{ab} = \sum_i Z_i (x_{ia} - c_a)(x_{ib} - c_b).$$

The weights make the frame element sensitive and reduce the contribution from hydrogen relative to heavier atoms. Scalar addends are sorted before summation, so floating-point accumulation is exactly independent of input atom order. Let $\lambda_1 \le \lambda_2 \le \lambda_3$ and V be the eigenvalues and orthonormal eigenvectors of T, and define

$$g_1 = (\lambda_2 - \lambda_1)/\lambda_3, \qquad g_2 = (\lambda_3 - \lambda_2)/\lambda_3.$$

When both gaps are at least 0.05, the ordered eigenvectors supply the frame. The threshold limits orientation amplification near an eigenspace degeneracy; it is a fixed conditioning policy, not a universal physical tolerance. If exactly one gap is smaller, the isolated eigenvector is retained. The ambiguous orthogonal plane is anchored by a centered atom vector projected into it. The atom is chosen by the lexicographically largest invariant integer key

$$\big(\mathrm{rint}(\|\mathbf{x}_i^{\perp}\|/\varepsilon), Z_i, \mathrm{rint}(|x_i^{\parallel}|/\varepsilon)\big).$$

Every atom tied for that key is evaluated, and the lexicographically smallest complete coordinate-row list wins. Thus ideal and near-planar benzene retain their unique normal axis while an atom anchors the molecular plane.

If both gaps are smaller, a first atom is selected by quantized radius and atomic number. A second, noncollinear atom is selected by projected radius, atomic number, and absolute axial coordinate. Again, every tie is evaluated. This ordered pair supplies a frame for spherical tops such as methane and for asymmetric point sets whose tensor is accidentally isotropic. Point-like inputs on the selected grid serialize zero coordinates. Exactly linear inputs use only their intrinsic axial coordinate, leaving the two meaningless transverse coordinates zero. These intrinsic representations avoid inventing unstable axes.

For every candidate frame, the centered coordinates are projected and quantized,

$$\mathbf{u}_i = \mathrm{rint}\Big((\mathbf{x}_i - \mathbf{c})^T V_* \, 10^p\Big).$$

All eight independent axis-sign combinations are evaluated. Rows $(Z_i, u_{i1}, u_{i2}, u_{i3})$ are sorted, and the lexicographically smallest complete row list is serialized. Atom-derived axes must extend at least ten coordinate-grid units, and no more than 10^4 tied frame candidates are evaluated. Failure of either conditioning rule issues a warning and invokes the canonical distance method; the result is visibly tagged `C` or `W`. No random perturbation, atom index, or retry limit affects the descriptor.

For example, water at $\varepsilon = 10^{-4}$ Å uses its planar anchored frame:

```
V:6-FRAME-SHA256|P:1.0e-04|Z:1,1,8|
F:1:0,-4688,-7572;1:0,-4688,7572;8:0,1172,0
```

Its identifier is

$$\texttt{H2Oq0m1-9a3a21fa2c3b6f0d4cb8acb76a18eccf}.$$

The same input under `method="canonical"` instead has a `C` body and suffix beginning `59781410`. These hashes are intentionally not comparable; a corpus must record the requested method and returned path.

3.5 Hashing and the readable identifier

The UTF-8 descriptor is hashed with SHA-256 [36], and the lowercase hexadecimal digest is truncated to L characters (4 bits each). The version, precision, atomic-number order, path tag, and path-specific body are all inside the hashed payload. Integer serialization removes differences in floating-point text formatting after quantization. A descriptor version change therefore changes the digest. The final identifier adds a readable prefix:

$$\underbrace{\texttt{H2O}}_{\text{Hill formula}}\ \underbrace{\texttt{q0m1}}_{\text{charge/spin tag}}\ \texttt{-}\ \underbrace{\texttt{9a3a21fa2c3b6f0d4cb8acb76a18eccf}}_{\text{geometry hash (32 hex, default)}}.$$

The Hill formula lists carbon first when present, then hydrogen, followed by other elements alphabetically; counts of one are omitted. The tag `q<charge>m<multiplicity>` records total charge and spin multiplicity, with a sign shown only for negative charge. If multiplicity is absent, the implementation assumes a singlet for an even electron count and a doublet for an odd count. This inference is only a metadata convenience: it does not identify triplet or higher-spin states of even-electron systems, nor higher-spin states of odd-electron systems, so callers must supply the multiplicity explicitly whenever the electronic state matters. Charge and multiplicity are not included in the hashed descriptor. Different electronic states of the same geometry therefore share a suffix, and users should not rely on suffix matching alone to distinguish state-specific energies, wavefunctions, or response properties.

3.6 Hash length and precision

For n distinct descriptors and $b = 4L$ bits, and in the sparse regime $n \ll 2^b$ for which hash lengths are chosen, the expected number of birthday collisions is approximately $n(n-1)/2^{b+1}$, independent of atom count N; the exact expected occupancy [Eq. (2)] governs the saturated regime. The default $L = 32$ (128 bits) keeps this expectation below one for $n < 2^{64.5} \simeq 2.6 \times 10^{19}$ distinct descriptors and below 10^{-9} for $n \lesssim 8.2 \times 10^{14}$. The helper `hash_length_for(n, p)` returns the minimum $L \in [1, 64]$ implied by the birthday approximation for target probability p (Table 2). If the calculation requires more than the 64 characters available from SHA-256, the helper returns 64; it does not report that the target is unattainable. Callers in this extreme regime must evaluate the probability separately or use a longer digest. This size-independent design parallels the approximately 65-bit skeleton block of the InChIKey [3, 37].

Precision sets the scalar grid spacing: canonical-frame coordinate components on `F`, or pairwise distances on `C`. A quantized scalar is stable under perturbations smaller than its margin to the nearest boundary, at most $\varepsilon/2$. Two values assigned to the same bin differ by at most ε, but two arbitrarily close values can occupy adjacent bins. On the canonical path, if every element-preserving atom matching contains a pair whose distances differ by more than ε, the descriptors must differ; smaller changes may or may not change them. The API accepts only $\varepsilon \in \{1, 10^{-1}, 10^{-2}, \ldots\}$ Å, which makes the `P` field and integer scale 10^p unambiguous. Invalid, nonfinite, Boolean, or excessively fine values are rejected before construction.

distinct geometries n	$p = 10^{-6}$	$p = 10^{-9}$	$p = 10^{-12}$
10^6	15	18	20
10^9	20	23	25
10^{12}	25	28	30
10^{15}	30	33	35

Table 2: Recommended geometry-hash length (hex characters) from `hash_length_for(`n`, `p`)`, keeping the birthday-collision probability below p for a namespace of n distinct geometries. The fixed default of 32 covers up to $\sim 8 \times 10^{14}$ items at $p = 10^{-9}$. The required length grows with corpus size, not with molecule size.

3.7 Implementation and complexity

The implementation depends only on NumPy and the Python standard library. The canonical method stores the $O(N^2)$ distance matrix. Each refinement round costs $O(N^2 \log N)$ for row sorting and lexicographic ranking. A generic geometry requires one round and one leaf, giving typical $O(N^2 \log N)$ time. Symmetry multiplies this cost by the number of search-tree nodes, usually of the order of the point-group size. On an Apple M1 Max laptop with one core, Python 3.10, and NumPy 2.2, small generic molecules take less than 1 ms, benzene about 1.4 ms, cubane about 7 ms, and a random 1000-atom cloud about 0.17 s.

An ideal monoelemental n-membered ring has a $(3n + 1)$-node tree and $O(N^3 \log N)$ total cost. Measured times are about 0.3 s for $n = 120$ (361 nodes), 7 s for $n = 360$, and 145 s for $n = 1000$. A degenerate-rounding input visits the full 10^4-node budget before fallback, taking about 1 s at $N = 50$ and 7 s at $N = 200$. Automorphism pruning could reduce this branching while preserving the lexicographic minimum. Section 5 reports routine-case measurements. The package also provides a command-line interface and an XYZ reader that validates the declared atom count and rejects malformed input.

On the regular frame branch, centroid and tensor evaluation is $O(N)$, the 3×3 eigendecomposition has constant cost, and row sorting gives $O(N \log N)$ time and $O(N)$ memory. An anchored branch evaluates A tied candidates and costs $O(AN \log N)$; A is normally one or a small symmetry orbit and is capped at 10^4. Point and line branches retain linear memory. Only an ill-conditioned anchor or candidate-budget overflow invokes the canonical method and its $O(N^2)$ memory requirement. On the full public corpus, isolated hash calls totaled 211 s (7,248 geometries/s); end-to-end iteration, diagnostic branch classification, and hashing totaled 297 s on one Apple M1 Max process. These are harness timings, not a cross-implementation speed claim.

3.8 Software-assisted development

Version 0.9.0 of the implementation was evaluated at commit `a23cf62`. Large language models—OpenAI GPT-5.6 (OpenAI, San Francisco, CA) and Anthropic Claude Opus 4.8 and Claude Fable 5 (Anthropic, San Francisco, CA)—were used to assist code implementation and debugging, analysis-script development, literature searches, and manuscript editing. They were used to accelerate these tasks, not as sources of numerical data or scientific authority. The author selected the methods, inspected the code, executed the calculations, checked the cited primary sources, and accepts responsibility for all results. Deterministic tests and released analysis artifacts provide the verification described below.

4 Invariance and Completeness Properties

Proposition 1 (Canonical-path invariance). *The canonical-path geometry hash is invariant, in exact arithmetic, under (a) any permutation $\pi \in S_N$ applied jointly to Z and $\mathbf{X}$, (b) any translation $\mathbf{x}_i \mapsto \mathbf{x}_i + \mathbf{t}$, (c) any orthogonal transformation $\mathbf{x}_i \mapsto R\mathbf{x}_i$ with $R \in O(3)$ (rotations, reflections, and inversion), and (d) any coordinate perturbation that does not change any quantized distance q_{ij}.*

Proof. Translation preserves coordinate differences. For $R^\top R = I$, $\|R(\mathbf{x}_i - \mathbf{x}_j)\| = \|\mathbf{x}_i - \mathbf{x}_j\|$, including when $\det R = -1$. Condition (d) preserves $\mathbf{Q}$ by definition.

The remaining construction is equivariant under atom relabeling. Initial colors depend only on Z; refinement uses sorted signatures and lexicographic ranks; and the target class is selected by its size and color. The search individualizes every member of that class. Relabeled inputs therefore produce isomorphic search trees with the same leaf strings. Their lexicographic minima are equal. The node-limit trigger depends only on the tree, and the fallback multiset is sorted. Thus both paths produce the same descriptor, and hence the same digest, under all stated operations. □

Because case (c) includes $\det R = -1$, enantiomers receive the same identifier. Applications that distinguish enantiomers must store a separate stereochemical tag.

Propositions 1–2 concern the mathematical descriptor. In the implementation, pairwise distances are evaluated in floating point, so an orthogonal transformation can alter the final unit in the last place and move a distance that lies within one ulp of a bin boundary across it. Exact invariance is therefore a property of the real-arithmetic descriptor; the portability requirements of Section 6 bound the gap between it and its finite-precision evaluation.

Proposition 2 (Completeness at the working precision). *On the canonical path, two labeled geometries receive the same descriptor if and only if there is a bijection π of their atoms that matches atomic numbers ($Z_i = Z'_{\pi(i)}$) and every quantized distance ($q_{ij} = q'_{\pi(i)\pi(j)}$ for all i, j). Consequently, barring a truncated-SHA-256 collision, two geometries with the same element-tagged distance multiset but nonisomorphic quantized distance matrices receive distinct hashes.*

Proof. ($\Leftarrow$) The proof of Proposition 1 establishes that the descriptor is a function of the pair $(Z, \mathbf{Q})$ considered up to joint relabeling; the hypothesized bijection says precisely that the two inputs share the same $(Z, \mathbf{Q})$ up to relabeling. ($\Rightarrow$) Equal descriptors have equal `Z` and `C` fields, so the two canonical orders σ, τ realize the same labeled matrix: $Z_{\sigma(k)} = Z'_{\tau(k)}$ and $q_{\sigma(k)\sigma(l)} = q'_{\tau(k)\tau(l)}$ for all k, l; then $\pi = \tau \circ \sigma^{-1}$ is the required bijection. □

Exactly congruent geometries have the same true and quantized distances. Equal descriptors provide the converse finite-grid certificate: an atom pairing exists for which all $\binom{N}{2}$ true distances agree to within the grid, $|d_{ij} - d'_{ij}| \le \varepsilon = 10^{-p}$ Å. The bound is inclusive because round-half-to-even can place both half-grid endpoints of an even bin—whose separation is exactly ε—in that bin. Exact distance equality implies congruence by the distance- geometry uniqueness theorem [22, 24]; at finite precision, however, this result bounds distances and does not imply an RMSD bound. Nor is the criterion "same hash if and only if within ε": nearby values can lie on opposite sides of a grid boundary. Moreover, proximity within ε is not transitive and cannot define an equivalence relation.

Proposition 3 (Deterministic frame-path properties). *For every accepted principal, intrinsic, or atom-anchored frame of Section 3.4, the* `F` *descriptor is invariant, in exact arithmetic, under joint atom relabeling, translation, every orthogonal transformation, and every perturbation that leaves its selected quantized coordinate-row multiset unchanged. Two* `F` *descriptors are equal if and only if these element-labeled row multisets are equal after minimization over all tied anchors and axis signs.*

A deterministic fallback tagged `C` *has Propositions 1–2; a returned* `W` *descriptor remains invariant but is not complete.*

Proof. Centering removes translations, and under an orthogonal transformation the gyration tensor transforms by orthogonal similarity. Isolated eigenspaces are therefore covariant. The anchor keys use only atomic numbers and quantized norms or absolute projections, which are invariant under relabeling and orthogonal transformation. Every tied atom or atom pair is evaluated, so a relabeling only permutes the candidate set. Taking its lexicographic minimum, then minimizing over all eight axis signs, removes candidate, sign, reflection, and input-order ambiguity. Point and line representations retain only intrinsic coordinates. Finally, equality of serialized `F` bodies is exactly equality of the retained (Z, u_1, u_2, u_3) row multisets. The fallback claims follow from the canonical construction and the distinct path tags. □

The frame path certifies coordinate rows in a deterministic canonical frame; the canonical path certifies matched pairwise distances. Their namespaces must not be mixed. The corpus, collision, adversarial-pair, and primary perturbation studies below use the default frame method; explicitly suffixed comparison artifacts report the retained canonical option.

Why multisets, and even atomic environments, are not enough. Flattening $\mathbf{Q}$ discards the assignment of distances to atom pairs. Consequently, a point set cannot always be reconstructed from its distance multiset [25, 26, 30]. Bloom's example consists of six points at $\{0, 1, 4, 10, 12, 17\}$ and $\{0, 1, 8, 11, 13, 17\}$ Å; the configurations share all 15 distances but are not congruent [32].

Retaining each atom's sorted, element-tagged distance environment separates this pair but remains incomplete. A randomized lattice search of 30,000 sets of 5–7 points from $\{0, \ldots, 3\}^3$ (`experiments/compare_algorithms.py`) found five-atom pairs with matching atomic environments but no matching distance matrix (Section 5). Related limitations have been established for invariant ML representations [15, 16], distance-based message passing [38], and pointwise distance distributions [39]. The canonical matrix separates all these examples.

Residual limits. Four qualifications apply. First, floating-point noise can move a distance or frame coordinate on a rounding boundary between adjacent bins. Second, a frame near a 0.05 gap threshold or an equality boundary between anchor keys can switch branches or anchors under perturbation; deterministic tie enumeration does not make the mapping continuous. Third, a search exceeding the 10^4-node budget uses the invariant but incomplete `W` fallback. An ideal monoelemental ring remains below the budget up to approximately 3000 atoms, but other highly degenerate integer matrices can exceed it at much smaller sizes and incur the full search cost before fallback. Fourth, truncation introduces the birthday-collision probability described below.

Truncation and birthday collisions. For $b = 4L$ bits and n distinct descriptors in the sparse regime $n \ll 2^b$, the expected number of accidental collisions is $n(n-1)/2^{b+1}$; the exact expected occupancy [Eq. (2)] holds when the table is not sparse. At the 128-bit default, one billion geometries give an expectation of 1.5×10^{-21}. The helper `hash_length_for` sizes L for a specified corpus and probability (Table 2); the same trade-off has been analyzed for the InChIKey [37].

5 Verification and Empirical Results

At version 0.9.0 (commit `a23cf62`), the package ships a test suite of 114 tests, all of which pass with Python 3.10 and NumPy 2.2.6. They cover the stated invariances; homometric and identical-

Check	Result
Default call	`F` descriptor, equal to explicit frame request
Rotation invariance (100 random proper rotations, water)	hash unchanged
Translation invariance (100 random translations)	hash unchanged
Permutation invariance (100 random relabelings)	hash unchanged
Reflection / parity invariance	hash unchanged
Enantiomer pair (CHFClBr and its mirror image)	identical hash (by design)
Linear molecule (CO_2) under rigid motions	hash unchanged
Benzene (D_{6h}), 50 joint relabelings + rotations	hash unchanged
Methane spherical top, 50 joint relabelings + rotations	anchored `F` hash unchanged
Explicit canonical option	retained `C` path in distinct namespace
Ill-conditioned frame	deterministic complete `C` fallback
Overlapping atoms (zero distance)	handled without error (not rejected; validation is left to the caller)
Isotopes	identical by construction (the API takes no masses)
Distinct bond lengths (H_2 at 0.74 vs 0.80 Å)	different hashes
Sub-precision noise (10^{-9} Å, $\varepsilon = 10^{-4}$)	hash unchanged
Charge and multiplicity varied	geometry hash unchanged
Determinism (repeated calls)	identical
Homometric pair (Bloom's 6-mark rulers, see below)	**separated**
Congruent mirror pair ($\{0, 1, 4, 6\}$ vs $\{0, 2, 5, 6\}$)	merged (by design)
Identical-environment 5-point pair (see below)	**separated**
Degenerate-rounding fallback (12-atom cluster at $\varepsilon = 1$ Å)	deterministic, invariant

Table 3: Independent verification of the default frame HashMol3D method and retained canonical option: 26 runtime checks, grouped into the 21 rows shown (the three counterexample rows each also assert their premises). The isotope result is definitional because the API has no mass input.

environment counterexamples; symmetric canonicalization; fallback; identifier format and length; input, XYZ, and command-line validation; power-of-ten precision; both public methods; and all frame-degeneracy branches. An independent harness uses the default frame method and explicitly checks the canonical option where path behavior matters; all 26 checks pass. Table 3 groups them into 21 rows. The homometric, congruent-mirror, and identical-environment rows each include assertions that establish the premise of the test as well as the expected identifier relation.

Homometric pairs are separated; congruent look-alikes are merged. For Bloom's pair [32], six carbon atoms lie at $A = \{0, 1, 4, 10, 12, 17\}$ or $B = \{0, 1, 8, 11, 13, 17\}$ Å. The configurations share all 15 distances, but exhaustive testing of all 6! atom pairings finds no matching distance matrices. A multiset descriptor merges them, whereas HashMol3D separates them. As a congruent control, the four-mark sets $A' = \{0, 1, 4, 6\}$ and $B' = \{0, 2, 5, 6\}$ share the multiset $\{1, 2, 3, 4, 5, 6\}$, but $B' = 6 - A'$ is a mirror image realizable by a proper rotation in 3D. HashMol3D therefore merges this pair, as required.

Beyond multisets: identical-environment pairs. Per-atom distance environments separate Bloom's pair but not the five-atom structures with carbon positions

$$A = \{(0, 2, 1), (1, 3, 3), (1, 1, 1), (3, 1, 1), (3, 3, 3)\},$$
$$B = \{(2, 0, 1), (3, 3, 1), (1, 1, 1), (1, 1, 3), (3, 3, 3)\}.$$

Each atom in one structure has an identical environment in the other, but none of the $5! = 120$ pairings matches the distance matrices. Environment descriptors merge these configurations [16, 38]; both the default frame and canonical descriptors separate them.

Rounding-boundary sensitivity. A distance on a rounding boundary (e.g. $xxxxx5$ at four decimals) can move between bins under floating-point noise. Bond lengths placed on opposite sides of such a boundary produce different hashes. A grid coarser than the input noise reduces the frequency of crossings but cannot guarantee stability for every geometry.

Coincident and near-coincident atoms. HashMol3D does not assess chemical validity. Two sites of a valid element that lie within half a grid unit yield $q_{ij} = 0$ and a deterministic identifier; note that $q_{ij} = 0$ certifies a separation below $\varepsilon/2$, not exact coincidence. This tolerance is useful when duplicate or overlapping atoms of a real element arise from format conversion, symmetry expansion, or united-atom and coarse-grained beads. Genuinely massless sites—ghost atoms in counterpoise corrections, dummy centers, and point charges—carry no atomic number in $\{1, \ldots, 118\}$ and are therefore outside the input domain; a caller that must hash them has to assign a placeholder element or screen them out first. Because the appropriate minimum separation depends on the application and precision, validation is left to the caller.

5.1 Comparison with alternative constructions

The relevant comparison depends on the task. Distance multisets, sorted per-atom distance environments, Coulomb-matrix eigenvalues [8], and many ML representations are invariant numerical summaries, but none is a complete invariant of a general element-labeled distance matrix. Continuous point-cloud invariants and metrics can instead support tolerance-based comparison and near-duplicate search [33–35]. HashMol3D addresses a narrower operation: exact lookup after a distance grid has been selected.

We tested three distance-based constructions on seven deliberately difficult noncongruent pairs: five homometric pairs and two pairs with identical per-atom distance environments. The distance multiset separated 0 of 7, the per-atom environment multiset separated 5 of 7, and the canonical matrix separated all 7. These examples establish the known failure modes of the two reductions; they are not a representative accuracy benchmark. Coulomb-matrix eigenvalues also separated these seven examples, but this finite result does not make the spectrum complete.

A naive principal-axis coordinate descriptor was fast for asymmetric inputs but failed rigid-motion invariance tests for benzene, methane, cubane, ideal rings, and a near-symmetric geometry because degenerate or nearly degenerate axes can change orientation. These failures motivated sign enumeration, intrinsic point/line representations, and canonical atom anchors. The resulting `F` path handles all of those examples without random shifts; only ill-conditioned anchors or a candidate-budget overflow invoke canonicalization. Reference timings and the complete comparison protocol are provided in the supplementary material.

5.2 Large-scale collision study on public datasets

We hashed 133,885 equilibrium geometries from QM9 ($N \in [3, 29]$; H, C, N, O, and F) [40] and complete MD17 trajectories [41] for aspirin (211,762 conformers, $N = 21$), benzene (627,983, $N = 12$), and ethanol (555,092, $N = 9$). QM9 samples different compositions and sizes, whereas MD17 contains many nearby conformers of each molecule. The combined set contains 1,528,722 geometries.

dataset	geometries	principal	one-axis	two-atom	line	`C` fallback
QM9	133,885	124,984	8,882	12	5	2
MD17 aspirin	211,762	211,762	0	0	0	0
MD17 benzene	627,983	177,361	450,622	0	0	0
MD17 ethanol	555,092	554,966	126	0	0	0
total	1,528,722	1,069,073	459,630	12	5	2

Table 4: Observed path decisions for the default frame method at $\varepsilon = 10^{-4}$ Å. "One-axis" retains an isolated tensor axis and anchors its degenerate plane; "two-atom" resolves a fully near-degenerate tensor. The two fallbacks were complete `C` descriptors; no `W` descriptor occurred.

			truncation-collision excess			
ε (Å)	n_ε	M_ε	24 bit	32 bit	40 bit	48–128 bit
1	312,179	1,216,543	2,900	10	0	0
10^{-1}	1,163,050	365,672	39,521	157	0	0
10^{-2}	1,527,923	799	67,791	269	3	0
10^{-3}	1,528,697	25	67,552	262	1	0
10^{-4}	1,528,722	0	67,484	266	0	0
10^{-5}	1,528,722	0	67,297	282	2	0

Table 5: Corpus precision sweep for the default frame request. The precision-merge excess M_ε counts repeated full digests among all $N_{\text{corpus}} = 1{,}528{,}722$ inputs. Truncation columns are calculated only after deduplicating those full digests, so they measure namespace truncation rather than quantization-induced coalescence. All tested lengths from 48 through 128 bits had zero truncation collisions.

Here a collision means that distinct full digests share a truncated prefix; repeated full digests could instead represent equal descriptors, and the streaming analysis does not retain bodies to distinguish that event from a full SHA-256 collision. Of the 1,528,722 default-method requests, 1,528,720 returned complete `F` descriptors, two QM9 structures used the complete `C` fallback, and none returned the incomplete `W` path (Table 4). All full 256-bit digests were distinct, as were their 128-bit prefixes. This is consistent with the negligible expected count at 128 bits; it is not empirical proof that SHA-256 truncation is collision free.

To separate the effect of grid resolution from the effect of digest length, we repeated the complete corpus calculation at all six accepted grids from 1 to 10^{-5} Å. Let n_ε be the number of distinct full 256-bit digests at grid ε, and define the full-digest repetition count

$$M_\varepsilon = N_{\text{corpus}} - n_\varepsilon. \tag{1}$$

The streaming calculation does not retain descriptor bodies, so equality of full digests is formally either equality of the quantized descriptors or a full SHA-256 collision. The latter is negligible at this scale; we therefore refer to M_ε as the *precision-merge excess*. It counts items beyond one per occupied full-digest value, not colliding pairs, and it is distinct from the truncated-prefix collision statistic below.

At the default 10^{-4} Å grid and the finer 10^{-5} Å grid, $M_\varepsilon = 0$. Coarsening to 10^{-3} Å produced 25 repetitions, all in QM9, a corpus fraction of 1.64×10^{-5}. At 10^{-2} Å, 135 QM9 structures and 664 benzene frames were excess items; aspirin and ethanol remained fully distinct. The behavior changes sharply at 10^{-1} Å: 365,672 inputs (23.92%) coalesced, including 45.56% of the densely sampled benzene trajectory. At 1 Å, 79.58% of the combined corpus coalesced, with 99.22% of benzene and 98.72% of ethanol frames in repeated-digest groups. The per-dataset counts are given in the supplementary material.

These data illustrate why precision is an application-level resolution choice rather than a universal collision-probability parameter. QM9 is separated substantially by composition and equilibrium structure, whereas neighboring frames in a fixed-composition MD trajectory can enter the same coarse grid cell. The result also need not be monotone across a path transition: at 1 Å every request used the complete `C` descriptor because the frame-conditioning rule requires a molecular span of at least ten grid units; from 10^{-1} through 10^{-4} Å, 1,528,720 requests used `F` and two used `C`, while at 10^{-5} Å one used `C`. No grid produced an incomplete `W` descriptor.

After full-digest deduplication, the truncation counts retain their ordinary namespace interpretation. Counts at 24 and 32 bits track the occupancy model for the corresponding n_ε; only 0–3 excess items occurred at 40 bits and none at any tested length from 48 through 128 bits (Table 5). Because the serialized precision field changes the SHA-256 input, prefix assignments are reshuffled between grids; small differences such as zero versus two 40-bit collisions are therefore not a monotone measure of geometric resolution.

Truncating the digest to b bits does produce collisions. Their number is set not by the sparse birthday form $n(n-1)/2^{b+1}$ but by the exact expected occupancy of a 2^b-slot table filled by n uniform digests—the number of items in excess of the occupied slots,

$$\mathbb{E}[\text{collisions}] = n - 2^b\Big[\,1 - \Big(1 - 2^{-b}\Big)^n\Big]\,. \tag{2}$$

Equation (2) reduces to $n(n-1)/2^{b+1}$ in the sparse limit $n \ll 2^b$; once the table saturates, the sparse form fails badly. At $b = 16$ the $n = 1.5 \times 10^6$ digests overflow the 65,536 available slots, and the sparse form predicts 1.8×10^7 collisions—more than the number of items—whereas Eq. (2) and the observed count agree at 1.46×10^6. Across the tested lengths, the counts are statistically consistent with Eq. (2): 67,484 collisions at 24 bits (6.76×10^4 expected), 266 at 32 bits (272 expected), and none at 40 bits or longer (Fig. 2).

5.3 Finite-grid sensitivity

A cryptographic identifier is discontinuous: changing one quantized scalar changes the descriptor and, with overwhelming probability, its digest. On the default frame path, those scalars are the $3N$ canonical-frame coordinate components. Their perturbations are correlated through the centroid, tensor, and selected frame; crossing a coordinate boundary can change a row, whereas crossing a gap or anchor-key boundary can change the frame itself. On the canonical path, the corresponding events are crossings among $\binom{N}{2}$ distance bins. Neither representation defines a continuous metric, and a fixed geometry need not have uniformly distributed bin phases.

We measured the default frame method for ten molecules at $\varepsilon = 10^{-4}$ Å, using 500 independent perturbations at each noise level (Fig. 3). At $\sigma = 0.01\varepsilon$, changes ranged from 0 for three panel members to 84.0% for cubane; at $\sigma = 0.03\varepsilon$, ethene changed in 15.8% of trials and $C_{16}H_{18}$ in 94.2%. Every system changed in at least 99% of trials by $\sigma = 0.3\varepsilon$. Geometry, atom count, frame conditioning, and boundary margins all matter, so these results do not define a molecule- independent safe noise level. The explicitly retained canonical comparison is reported in `flip_sweep_canonical.csv`; its ordering differs, but it is likewise boundary sensitive.

Exact rigid transformations and atom permutations changed none of 210 tested frame descriptors for each operation. The test used ten unique molecules; writing and reading coordinates at six decimal places changed 2 of 10 frame descriptors, whereas four- and three-decimal storage changed

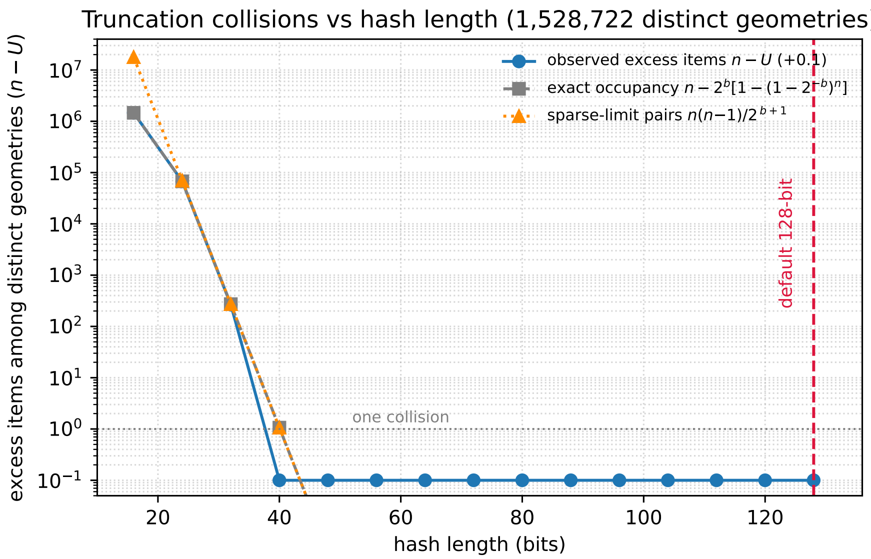

Figure 2: Truncation collisions as a function of hash length. Observed counts (blue circles) are consistent with the exact expected occupancy [Eq. (2), gray dashed]. The sparse birthday form $n(n-1)/2^{b+1}$ (orange dotted) is valid only in the sparse tail and overshoots sharply once the table saturates ($b \leq 24$). Values are offset by 0.1 to show zeros on the logarithmic axis; the dashed vertical line marks the 128-bit default.

all 10. Repeating each deterministic round trip after different rigid transformations does not create independent file-format samples, so these counts are reported per unique molecule. Gaussian noise with $\sigma = \varepsilon/10$ changed 204 of 210 frame descriptors. The canonical option gave 0 of 210 rigid/permutation changes, 3 of 10 six-decimal changes, and 198 of 210 noise changes.

Thus, storing two more coordinate decimals than the grid exponent is not a sufficient general rule. Coarser quantization reduces, but cannot eliminate, boundary sensitivity. Users who need tolerance-based matching should compare structures with an alignment or continuous invariant rather than expect an exact hash to supply a metric. The supplementary material reports the full protocol and exploratory precision studies.

6 Limitations

Reflection-related structures, including enantiomers, share an identifier. Stereochemistry must be stored separately when required. Atomic labels contain only the atomic number, so isotopologues also share an identifier. Charge and spin appear in the readable prefix rather than the hashed descriptor; the hash suffix alone does not specify electronic state. The identifier is not invertible and cannot recover Cartesian coordinates.

Completeness is path specific. The `F` path certifies its element-labeled canonical-frame rows (Proposition 3); the `C` path certifies isomorphism of the element-labeled quantized matrix (Z, Q). Neither is an ε-ball relation on Cartesian coordinates, and coarse quantization can merge noncongruent structures. Empirical claims must therefore record the path and exclude the incomplete `W` fallback. Frame coordinates, distances, gap tests, and anchor keys all have boundaries at which floating-point noise can change the descriptor. Inputs that exceed the canonical search budget use the invariant, deterministic, but incomplete `W` path. Accepted precisions are powers of ten no coarser

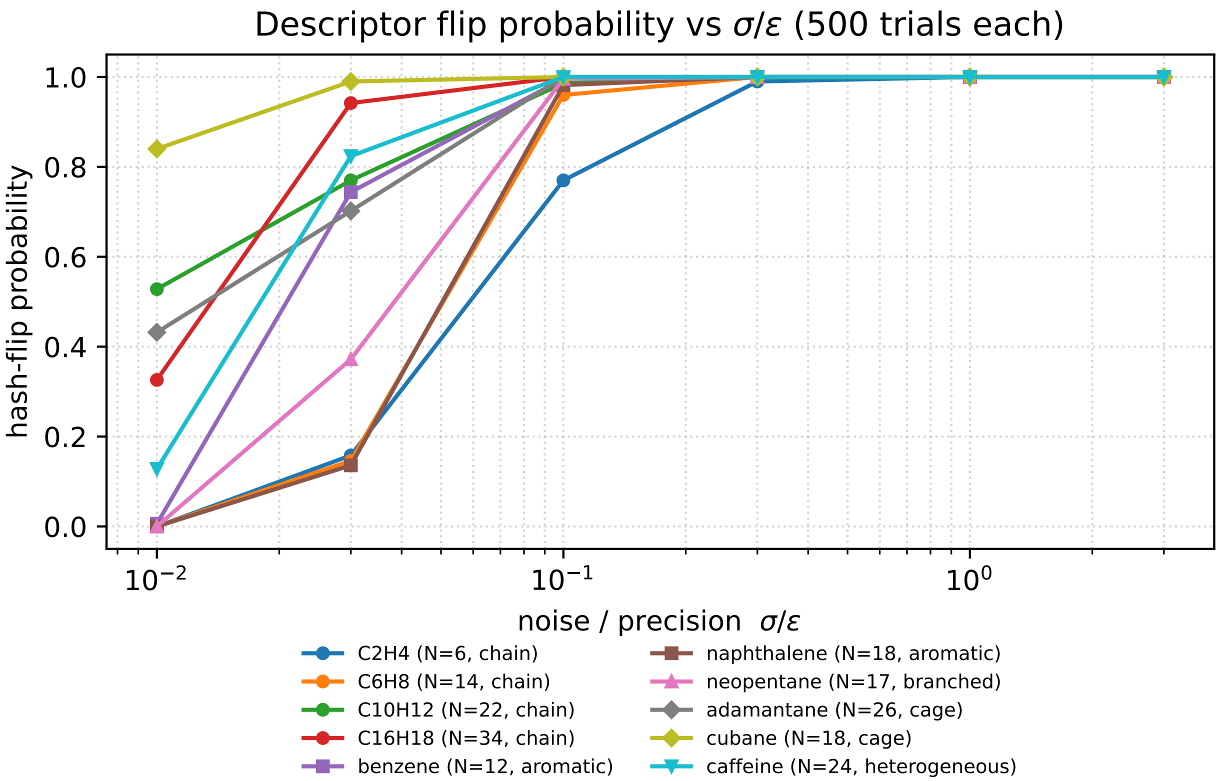

Figure 3: Fraction of default frame descriptors changed by isotropic per-atom Gaussian noise at $\varepsilon = 10^{-4}$ Å (500 trials per point). The abscissa is σ/ε. Curves correspond to ten molecules. The transition depends on geometry, atom count, frame conditioning, and coordinate- grid boundary margins.

than 1 Å.

The canonical method requires $O(N^2)$ memory. Symmetry multiplies its $O(N^2 \log N)$ refinement cost by the search-tree size; ideal monoelemental rings reach $O(N^3 \log N)$ and take minutes at $n = 1000$ (Section 3). The frame method uses $O(N)$ memory for principal, intrinsic, and accepted anchored structures, but fallback incurs the canonical cost. Its `F` hashes are not comparable with `C` or `W` hashes. Neither method supports periodic boundary conditions.

Cross-platform reproducibility requires the same descriptor version, precision, quantization rule, node budget, and hash function. Frame calculations also require the same requested method, principal-gap threshold, anchor conditioning threshold, and candidate budget. Although the specification fixes these values, a distance or frame coordinate within one ulp of a grid boundary could round differently across platforms (Sections 3.3 and 3.4).

7 Applications

Potential applications include conformer deduplication before QC refinement—by exact equality of the quantized descriptor, not tolerance-based matching [42, 43]; caches for scalar energies keyed by the identifier *together with* the electronic state and computational model; provenance tracking in QC and ML datasets [40]; indexing molecular-dynamics frames; and stable structure references in automated workflows. The prefix displays formula, charge, and spin, while the suffix identifies the quantized geometry. Caching gradients or other tensorial quantities additionally requires the atom permutation and orthogonal transformation relating the cached and query frames, because such quantities transform under relabeling and rotation; the scalar identifier alone does not supply that mapping. Because reflections and isotopes share an identifier, applications sensitive to chirality or isotopic substitution must store the distinguishing information separately.

8 Conclusion

HashMol3D provides a portable identifier for a quantized 3D molecular geometry. The default descriptor uses $O(N)$ memory and a deterministic frame: principal axes for separated moments, intrinsic point/line coordinates, and canonical atom anchors for degenerate eigenspaces. It is invariant under translation, orthogonal transformations, and atom relabeling. The complete canonicalized distance matrix remains available as `method="canonical"` and as the conditioned fallback. Tests separate homometric and identical- environment counterexamples, merge a congruent mirror pair, and observe no 128-bit collisions among 1,528,722 QM9 and MD17 geometries; only two corpus items required canonical fallback and none used `W` at the default grid. Sweeping precision from 1 to 10^{-5} Å preserved every full digest at 10^{-4} and 10^{-5} Å, but increasingly merged nearby trajectory frames at coarser resolutions. Possible extensions include stereochemical tags, periodic cells, automorphism pruning for extreme symmetries, and additional structure readers.

Supplementary Material

See the supplementary material for reference timings, exploratory polyene perturbations and local-curvature probes, conformer and quantum-chemistry workflow checks, the per-dataset corpus precision sweep, and the detailed boundary-margin and round-trip analyses.

Author Declarations

Conflict of Interest

The author has no conflicts of interest to disclose.

Author Contributions

Murat Keçeli: Conceptualization; methodology; software; validation; formal analysis; investigation; data curation; visualization; writing—original draft; writing—review and editing.

Data and Code Availability

The implementation, documentation, specification, and tests are available as HashMol3D version 0.9.0, commit `a23cf62`, at https://github.com/Autonomous-Scientific-Agents/HashMol3D under the MIT license. The manuscript source, supplementary material, analysis scripts, CSV data products, and committed figures are available at https://github.com/keceli/paper-hashmol3d. Corpus and primary perturbation studies use the default frame method; explicitly suffixed CSV/PDF artifacts report the retained canonical comparison. The algorithm-comparison harness is `experiments/compare_algorithms.py` in the implementation repository. The six-grid corpus sweep is generated by `analysis_precision_corpus.py`; its aggregated products are `precision_sweep.csv` and `precision_sweep_by_dataset.csv`.

The polyene, local-curvature, collision, and final perturbation artifacts used Python 3.13, NumPy 1.26.4, SciPy 1.14.1, RDKit 2024.09.6 [44], PySCF 2.9.0 [45], geomeTRIC 1.0.2 [46], and Matplotlib 3.9.3. These are optional analysis dependencies; the core package requires only NumPy, and its test suite was also validated under Python 3.10 and NumPy 2.2.6. Every stochastic step uses a fixed seed, although RDKit conformer generation can vary between library versions. The collision loader reads QM9 [40] and MD17 [41] from the sources and with the expected per-dataset geometry counts recorded in the manifest at the head of `analysis_collisions.py`, and it aborts on a missing or partial dataset so that a truncated download cannot silently change the reported totals. SHA-256 checksums for the four input files are recorded in `dataset_manifest.sha256`.

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